Fiscal Year:	FY 2005	**Task Last Updated:**	FY 06/17/2008
PI Name:	Maurer, Richard H. Ph.D.		
Project Title:	Combined Ion and Neutron Spectrometer for Space Applications		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Technology Development Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Laurel	**State:**	MD
Zip Code:	20723-6099	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Biomedical Research & Countermeasures 03-OBPR-04
Start Date:	08/01/2004	**End Date:**	07/31/2008
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Roth, David (Johns Hopkins Applied Physics Laboratory) Zeitlin, Cary (Lawrence Berkeley National laboratory) Goldsten, John (Johns Hopkins Applied Physics Laboratory) Haggerty, Dennis (Johns Hopkins Applied Physics Laboratory)		
Grant/Contract No.:	NCC 9-58-TD00404		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	We have begun to design and fabricate a prototype Combined Ion and Neutron Spectrometer (CINS) for Space Applications. This instrument will improve upon existing charged particle species and energy spectrometers (e.g., MARIE on the Mars Odyssey mission) and incorporate the capability of the previously developed NSBRI Neutron Energy Spectrometer to yield a single and complete ionizing radiation environment monitor for application in space habitats and transport vehicles. An improved charged particle/ion telescope detector system is being designed at Lawrence Berkeley Laboratory. It will be combined with the NSBRI Neutron Spectrometer at The Johns Hopkins University Applied Physics Laboratory into a single instrument for ground based research at accelerators by using multiple fast channels of analog and digital electronics based on JHU/APL's experience with NEAR and MESSENGER spacecraft instrument suites. This approach merges the sophistication of space flight design with the low cost of non-flight electronic components. After fabrication and calibration, CINS will be used to compare its energy spectra with the Tissue Equivalent Proportional Counter (TEPC) to investigate basic hypotheses about 1) the lack of radiation environment information from the TEPC LET (Linear Energy Transfer) spectrometer due to detection limitations with respect to high energy protons, low energy heavy ions and almost all neutrons; and 2) the increased risk due to the multiplication of primary beams of protons and heavy ions by spacecraft materials which produce a secondary particle environment of protons, neutrons and heavy ion fragments. This investigation will determine the limitations of present ionizing radiation dosimetry and develop correction factors that can be applied to TEPC and dosimeter measurements for a more accurate estimate of dose equivalent and risk to spaceflight personnel.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	A compact radiation environmental monitor may have application for homeland security.
Task Progress:	Since August 2004 we have been concentrating on the conceptual design of the detectors for the instrument and the fabrication/procurement of these solid state devices for charge collection or scintillation.
Bibliography Type:	Description: (Last Updated: 04/27/2011)
Articles in Peer-reviewed Journals	Kinnison JD, Maurer RH, Roth DR, Haight RC. "High-energy neutron spectroscopy with thick silicon detectors." Radiat Res. 2003 Feb;159(2):154-60. PMID: 12537520 , Feb-2003